Online Appendix for

"Correlated Country Risk and International Capital Allocation"

Siyoung Lee* Jun Hee Kwak†

Sogang University Sogang University

November 12, 2026

A Additional Empirical Results

A.1 Bilateral Portfolio Friction Estimation

Table A.1 reports the PPML estimates of bilateral portfolio frictions, including geographic distance, home bias, and geopolitical alignment.

*siyoungecon@gmail.com

†jhkwak@sogang.ac.kr

Table A.1: Bilateral Portfolio Frictions and Geographic Distance

	(1)
	share
$\ln(dist_{i,j})$	-0.797***
	(0.0544)
Foreign	-4.159***
	(0.130)
Cross Block	-0.796***
	(0.155)
Observations	100377

Standard errors in parentheses

Standard errors clustered by country-pair (country x counterpart).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.2 σ_N Estimation

The return term is endogenous because equilibrium interest rates are jointly determined with international portfolio allocation. To estimate σ_N, we use an IV strategy that builds on Koijen and Yogo (2026) and adapts their residual-supply approach to our international capital-allocation setting. The instrument exploits variation in predicted asset demand relative to predicted asset supply. This residual demand pressure shifts equilibrium returns through asset-market clearing but, under the exclusion restriction, does not directly affect investor n's bilateral portfolio demand. It therefore provides exogenous variation in returns that identifies the sensitivity of portfolio allocation to equilibrium returns.

Equation (3) can be modified to be expressed as

$$\log \frac{b_{ni}}{b_{nn}} = \frac{1}{\sigma_N}(r_{it} - r_{nt} - \Phi_{ni} + \omega_{it} - \omega_{nt})$$

Let X_{ni} denote the observed bilateral characteristics between investor country n and destination country i, including bilateral geographic distance and the cross-bloc indicator. The identification

assumptions for the IV strategy are:

$$\begin{aligned} \text{Exogeneity:} \quad & \operatorname{Cov}\left(\varepsilon_{nit},\, dist_{ji} \mid X_{ni}, \iota_i, \eta_n\right) = 0, \\ \text{Relevance:} \quad & \operatorname{Cov}\left(r_{it},\, dist_{ji} \mid X_{ni}, \iota_i, \eta_n\right) \neq 0, \qquad j \neq n. \end{aligned}$$

The exogeneity condition requires that, conditional on the bilateral characteristics between investor country n and destination country i, the geographic distance between other investor countries j and destination i is uncorrelated with the idiosyncratic portfolio-demand shock of investor n. Intuitively, the location of other investor countries should not directly affect country n's relative preference for destination i.

The relevance condition requires that the geographic distribution of other investor countries affects the equilibrium return in destination i. Because bilateral portfolio demand declines with geographic distance, destinations that are relatively close to countries with greater investment capacity face stronger predicted demand. Through capital-market clearing, this variation in demand pressure affects the equilibrium return r_{it}. We aggregate this cross-country distance variation into the residual-demand instrument constructed below.

Demand Component From equation (3), we construct the time-invariant component of portfolio demand using only the estimated bilateral gravity frictions. Specifically, the predicted allocation share from investor country m to destination country i is given by

$$\hat{b}_{mi}^{gravity} = \frac{\exp(-\phi_{mi}^{gravity})}{\sum_h \exp(-\phi_{mh}^{gravity})},$$

where

$$\phi_{ni}^{gravity} = \hat{\beta}^{dist} \ln(dist_{ni}) + \hat{\beta}^{foreign} \mathbb{1}[n \neq i].$$

This predicted share captures the time-invariant geographic structure of portfolio demand implied by bilateral distance and home bias, abstracting from time-varying destination characteristics and equilibrium returns. We then aggregate predicted demand from all investor countries other than n:

$$\sum_{m \neq n} A_{mt} \hat{b}_{mi}^{gravity}.$$

Supply Component To construct the supply component, we estimate the following panel regression:

$$\log S_{it} = \Psi' z_{it} + \lambda_t + \varepsilon_{it}^S,$$

where $z_{it} = [\log GDP_{it}, \log pop_{it}]'$, and λ_t denotes year fixed effects. We then construct predicted asset supply as

$$\log \hat{S}_{it} = \hat{\Psi}' z_{it} + \hat{\lambda}_t.$$

The predicted value captures the component of asset supply explained by observable country characteristics and common time variation.

Residual Demand Pressure Using the predicted asset allocation from the gravity model, we first construct a leave-one-out measure of demand pressure for each destination i:

$$Z_{i,-n,t} = \log \left(\sum_{m \neq n} A_{mt} \hat{b}_{mit}^{gravity} \right) - \log \hat{S}_{it}.$$

This measure is the negative of the conventional residual-supply measure. A larger value of $Z_{i,-n,t}$ therefore indicates stronger predicted demand relative to asset supply in destination country i. The leave-one-out construction excludes predicted demand originating from investor country n, so that the bilateral portfolio choice of investor n does not mechanically enter the demand component. Because the endogenous regressor is the return differential between destination i and country n, we use the corresponding difference in residual demand pressure as the instrument:

$$Z_{nit}^{rel} = Z_{i,-n,t} - Z_{n,-n,t}.$$

Thus, Z_{nit}^{rel} measures the relative predicted demand pressure on destination i compared with country n, generated by investors other than n. Variation in this relative demand pressure shifts the equilibrium return differential $r_{it} - r_{nt}$ through asset-market clearing.

2SLS The return elasticity of portfolio demand is estimated using two-stage least squares. Since the portfolio-demand equation is expressed in terms of returns relative to the investor's domestic asset, we instrument the return differential $r_{it} - r_{nt}$ with the corresponding difference in leave-one-out

residual demand pressure. The first stage is

$$r_{it} - r_{nt} = \pi Z^{rel}_{nit} + \mathbf{X}'_{ni}\beta + \nu_{nit},$$

where

$$Z^{rel}_{nit} = Z_{i,-n,t} - Z_{n,-n,t}.$$

The estimated return coefficient implies $\hat{\sigma}_N = \frac{1}{\hat{\lambda}_r} \approx 0.074$

Table A.2: IV Estimation Results

	(1)	(2)
	First stage	2SLS
Relative Z	0.01***	
	(0.00)	
ΔR		13.58***
		(3.39)
First-stage F (excl. instr.)		688.33
Kleibergen-Paap rk Wald F		688.33
Observations	28772	28772
Adjusted R^2		0.092

Fixed effects: country and counterpart. Robust standard errors in parentheses.

B Robustness Checks

We examine whether the estimated effects of geographic distance and geopolitical alignment on portfolio allocation are robust to alternative specifications. Recall that the bilateral wedge ϕ_{ni} enters portfolio demand with a negative sign. Thus, a negative coefficient in the portfolio allocation equation corresponds to a larger portfolio wedge.

First, we augment the baseline specification with additional bilateral covariates, including exchange-rate volatility, common language, and trade intensity. Table B.1 reports the baseline estimate in the full sample, the baseline estimate in the common sample, and specifications that sequentially add these controls. The coefficients on log geographic distance and the cross-block indicator remain

negative and statistically significant across specifications.

Table B.1: Robustness of Bilateral Portfolio Friction Estimates

	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(dist_i,j)$	-0.797***	-0.790***	-0.716***	-0.726***	-0.706***	-0.650***
	(0.054)	(0.055)	(0.052)	(0.058)	(0.063)	(0.062)
Foreign	-4.159***	-4.171***	-3.730***	-4.424***	-4.478***	-3.991***
	(0.130)	(0.131)	(0.117)	(0.144)	(0.156)	(0.148)
Cross Block	-0.796***	-0.803***	-0.705***	-0.773***	-0.819***	-0.722***
	(0.155)	(0.157)	(0.172)	(0.156)	(0.157)	(0.171)
FX volatility (%)			-0.288***			-0.279***
			(0.030)			(0.030)
Common language				0.661***		
				(0.169)		
Trade intensity (%)					0.141***	0.108***
					(0.039)	(0.036)
Sample	Full	Common	Common	Common	Common	Common
Observations	100377	95086	95086	95086	95086	95086

Standard errors in parentheses

Origin×year and destination×year fixed effects. Standard errors clustered by country pair.

FX volatility: s.d. of monthly bilateral exchange-rate returns within the year (%).

Trade intensity: (X_ij+X_ji)/(GDP_i+GDP_j) (%), IMF IMTS and World Bank.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We also allow home bias to vary across investor countries. Specifically, we parameterize the bilateral wedge as

$$\phi_{ni} = \beta^{dist} \ln(dist_{ni}) + \beta_n^{foreign} \mathbb{1}[n \neq i] + \beta^{cross\text{-}block} CrossBlock_{ni}.$$

Allowing for investor-specific home bias leaves the main results unchanged. The coefficients on log geographic distance and the cross-block indicator in the portfolio allocation equation remain negative and statistically significant.

Table B.2: Robustness to Investor-Specific Home Bias

	(1) share
Log distance	-0.921***
	(0.045)
Cross block	-0.579***
	(0.216)
Observations	100377.000

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Validation of Spatial Attractiveness

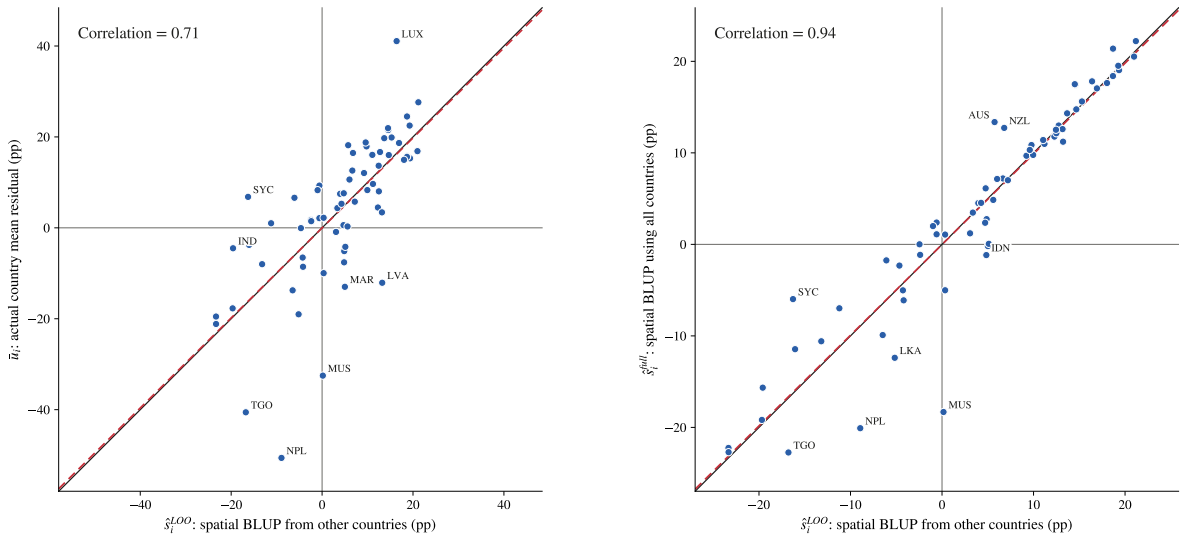
We assess the estimated spatial component of destination attractiveness using a leave-one-country-out (LOO) procedure. For each country, we construct a spatial BLUP using information from all other countries, excluding its own residual from the prediction.

Figure C.1 presents two validation exercises. Panel A compares the LOO spatial predictions with observed country-level mean residuals. The two measures are positively correlated (0.71), suggesting that the geographic structure of destination attractiveness allows country-level differences to be predicted using information from other countries.

Panel B compares the full-sample and LOO spatial BLUPs. The two estimates are highly correlated (0.94), indicating that the estimated spatial component is not primarily driven by each country's own residual but instead reflects geographic information shared across countries.

D Credit Risk and Spatial Dependence

This appendix provides additional evidence on the geographic structure of sovereign credit risk and its relationship with destination attractiveness. We examine whether sovereign credit risk exhibits persistent spatial dependence and whether its spatial component is associated with the spatial structure of destination attractiveness documented in the main text. Throughout the analysis, we measure country credit risk using sovereign credit-rating scores from Refinitiv Datastream, where higher values indicate greater creditworthiness and therefore lower credit risk.



(a) Prediction from Other Countries

(b) Full-Sample vs. LOO Spatial BLUPs

Figure C.1: Validation of Spatial Attractiveness

D.1 Geographic Patterns in Sovereign Credit Risk

We first examine whether bilateral differences in sovereign credit risk vary systematically with geographic distance. We consider two measures of sovereign credit risk, sovereign credit-rating scores and CDS spreads. For each country, we compute the average credit-rating score and average log CDS spread over 2012–2023. Following the empirical specification for interest-rate differentials and destination attractiveness, we estimate

$$|\bar{Z}_i^k - \bar{Z}_j^k| = \beta^k \ln(dist_{ij}) + \alpha_i^k + \alpha_j^k + \varepsilon_{ij}^k, \tag{D.1}$$

where $k \in \{C, CDS\}$ denotes the measure of sovereign credit risk. $\bar{Z}_i^C$ represents the average sovereign credit-rating score of country i, while $\bar{Z}_i^{CDS}$ denotes its average log CDS spread.

Table D.1 reports the estimation results. Geographic distance is positively and significantly associated with bilateral differences in both sovereign credit-rating scores and log CDS spreads. These findings indicate that geographically distant countries tend to exhibit larger differences in sovereign credit risk, consistent with the geographic patterns documented for destination attractiveness in the main text.

Table D.1: Geographic Distance and Sovereign Credit Risk Differences

	(1) Sov. score gap	(2) CDS gap
Log distance	0.317***	0.191***
	(0.030)	(0.024)
Origin FE	Yes	Yes
Destination FE	Yes	Yes
Countries	145	67
Observations	10,438	2,209
R^2	0.299	0.529

Robust standard errors in parentheses.
* p<0.10, ** p<0.05, *** p<0.01

D.2 Sovereign Credit Risk and Spatial Destination Attractiveness

The geographic patterns in sovereign credit risk documented above suggest that creditworthiness may be one economic characteristic underlying the spatial dependence in destination attractiveness. To examine this possibility, we jointly model the persistent components of sovereign creditworthiness and destination attractiveness. We focus on sovereign credit-rating scores, which provide broader country coverage than CDS spreads in the joint estimation sample. The rating score is increasing in sovereign creditworthiness. The joint estimation uses the overlapping 2012–2020 period.

D.2.1 Joint Spatial Covariance

Following the spatial decomposition of destination attractiveness in the main text, we specify the bivariate model

$$\begin{pmatrix} C_{it} \\ \omega_{it} \end{pmatrix} = \begin{pmatrix} \alpha_t^C \\ \alpha_t^\omega + \gamma_P \ln Pop_{it} \end{pmatrix} + \begin{pmatrix} s_i^C \\ s_i^\omega \end{pmatrix} + \begin{pmatrix} a_i^C \\ a_i^\omega \end{pmatrix} + \begin{pmatrix} \varepsilon_{it}^C \\ \varepsilon_{it}^\omega \end{pmatrix}, \tag{D.2}$$

where C_{it} denotes the sovereign credit-rating score, s_i^C and s_i^ω are persistent spatial components, and a_i^C and a_i^ω capture persistent non-spatial country heterogeneity. We allow the non-spatial country components to be correlated across the two outcomes.

We model the joint spatial covariance as

$$\operatorname{Cov}\left(\begin{pmatrix} s_i^C \\ s_i^\omega \end{pmatrix}, \begin{pmatrix} s_j^C \\ s_j^\omega \end{pmatrix}\right) = \exp\left[-\left(\frac{d_{ij}}{\rho}\right)^2\right] \underbrace{\begin{pmatrix} \sigma_{s,C}^2 & \sigma_{s,C\omega} \\ \sigma_{s,C\omega} & \sigma_{s,\omega}^2 \end{pmatrix}}_{\Sigma_s}, \tag{D.3}$$

where ρ is the common geographic correlation range. The parameter $\sigma_{s,C\omega}$ measures the spatial cross-covariance between sovereign creditworthiness and destination attractiveness. The corresponding spatial correlation is

$$r_s = \frac{\sigma_{s,C\omega}}{\sqrt{\sigma_{s,C}^2 \sigma_{s,\omega}^2}}. \tag{D.4}$$

We estimate the joint spatial covariance structure using restricted maximum likelihood. Table D.2 reports the results. The estimates indicate a positive spatial relationship between sovereign creditworthiness and destination attractiveness. This suggests that geographically correlated creditworthiness may be one economic characteristic underlying the persistent spatial dependence in destination attractiveness.

Table D.2: Joint Spatial Covariance of Sovereign Creditworthiness and Destination Attractiveness

Sample	Countries	$\hat{\sigma}_{s,C\omega}$	$\hat{r}_s$	LR	Bootstrap p
Baseline	65	0.238	1.000	5.21	0.025
Excluding Venezuela	64	0.242	1.000	13.99	0.005
At least 5 observations	59	0.251	0.999	8.38	0.005

Notes: The table reports joint spatial REML estimates for sovereign creditworthiness and destination attractiveness over 2012–2020. The likelihood-ratio test examines the null of zero spatial cross-covariance. Bootstrap p-values are based on 199 simulations. The minimum-observation restriction refers to destination attractiveness.

E Geographic and Geopolitical Dependence

Table 2b provides evidence that countries' estimated latent attractiveness is associated with both geographic and geopolitical proximity, with the latter measured by distance in UN General Assembly voting ideal points. To capture these two sources of dependence, we specify an additive Gaussian covariance kernel:

$$\operatorname{Cov}(s_i, s_j) = \sigma_G^2 \exp\left[-\left(\frac{\tilde{d}_{G,ij}}{\rho_G}\right)^2\right] + \sigma_P^2 \exp\left[-\left(\frac{\tilde{d}_{P,ij}}{\rho_P}\right)^2\right]. \tag{E.1}$$

Here, the subscripts G and P denote the geographic and geopolitical components, respectively. Geographic distance is measured by chordal distance, while geopolitical distance is the absolute difference between countries' sample-period average ideal points. Each distance is divided by its standard deviation across distinct unordered country pairs, without subtracting its mean. The parameters σ_G^2 and σ_P^2 govern the component variances, while ρ_G and ρ_P govern their correlation ranges.

This specification corresponds to the decomposition $s_i = s_i^G + s_i^P$, with $\mathrm{Cov}(s_i^G, s_j^P) = 0$ for all i, j. This restriction concerns the latent components and does not require geographic and geopolitical distances to be uncorrelated. We also retain the independent country-specific component a_i, with variance σ_a^2.

Using the geopolitical specification of ω_{it}, with 1,028 country-year observations from 67 countries and controlling for log population and year fixed effects, we estimate $\hat{\sigma}_G^2 = 0.00552$, $\hat{\sigma}_P^2 = 0.01501$, and $\hat{\sigma}_a^2 = 0.00791$. These components account for approximately 19.4%, 52.8%, and 27.8% of the model-implied persistent variance, respectively. The estimated decomposition therefore attributes a substantial share of persistent heterogeneity to geopolitical dependence, while retaining geographic dependence and an independent country-specific component.

Converted to the original distance units, the estimated geographic and geopolitical ranges are 1,677 km and 1.366 ideal-point units, respectively. At these distances, the correlation within the corresponding component falls to e^{-1}, approximately 0.37. The geographic range is shorter than the 2,929 km estimated under the geography-only covariance specification, indicating a reallocation of covariance when geopolitical dependence is modeled separately.

A parametric bootstrap test with 499 replications rejects the null hypothesis $H_0 : \sigma_P^2 = 0$ at the 5% level ($p = 0.012$). This result provides evidence of geopolitical dependence beyond that captured by the geographic covariance component.

F Destination Attractiveness and Country Characteristics

Table F.1 examines how the recovered destination attractiveness, $\omega_{i,t}$, is related to observable country characteristics. By construction, $\omega_{i,t}$ captures components of destination attractiveness other than interest rates.

Population is positively associated with destination attractiveness across countries, consistent with larger markets attracting greater portfolio demand. Sovereign creditworthiness is also positively and significantly associated with destination attractiveness across specifications, including those with country fixed effects. This suggests that countries with stronger sovereign credit ratings tend to attract greater portfolio demand conditional on observed returns.

Table F.1: Destination Attractiveness and Country Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)
ln(Population)	0.055***		0.059***	-0.439**		-0.406*
	(0.012)		(0.011)	(0.204)		(0.206)
Sovereign Credit Score		0.030***	0.034***		0.011**	0.008*
		(0.008)	(0.006)		(0.005)	(0.004)
Country FE	No	No	No	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	531	531	531	528	528	528
Countries	66	66	66	63	63	63
Adj. R^2	0.281	0.132	0.473	0.960	0.958	0.960

Standard errors clustered by country in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G Model Solution Algorithm

G.1 Solution Algorithm

We solve the stationary general equilibrium using a nested numerical procedure. All parameters, including the calibrated discount factors $\{\beta_n\}$, are held fixed throughout the solution. We use the normalized bilateral portfolio wedge

$$\phi_{ni} \equiv \frac{\Phi_{ni}}{\sigma_N},$$

as defined in Section 5.1.

For a given vector of asset supplies $\{A_n\}$, the inner loop solves for capital allocation, portfolio shares, and interest rates. Given a current guess for capital $\{k_i\}$, interest rates are

$$r_i = \alpha Z_i \left(\frac{k_i}{L_i}\right)^{\alpha-1} \tag{G.1}$$

Portfolio shares are then determined by

$$b_{ni} = \frac{\exp\left(-\phi_{ni} + \omega_i/\sigma_N + r_i/\sigma_N\right)}{\sum_j \exp\left(-\phi_{nj} + \omega_j/\sigma_N + r_j/\sigma_N\right)}, \tag{G.2}$$

with the modification for destinations outside the model sample described in Appendix G.2. The implied capital stock is

$$k_i' = \sum_n b_{ni} A_n. \tag{G.3}$$

We iterate over $\{k_i\}$ until the implied capital allocation converges.

Given the resulting portfolio allocation and interest rates, the outer loop adjusts asset supplies until the steady-state Euler condition

$$\beta_n R_n = 1 \tag{G.4}$$

is satisfied for every investor country, where

$$R_n = 1 + \sum_i b_{ni} r_i. \tag{G.5}$$

Algorithm 1 Solution of the Stationary General Equilibrium

Initialize asset supplies $\{A_n\}$.

repeat

  Initialize destination capital stocks $\{k_i\}$.

  repeat

    Compute interest rates:

$$r_i = \alpha Z_i \left(\frac{k_i}{L_i}\right)^{\alpha-1}$$

    Compute bilateral portfolio shares $\{b_{ni}\}$.

    Update destination capital:

$$k_i' = \sum_n b_{ni} A_n.$$

  until $\{k_i\}$ converges

  Compute investor portfolio returns:

$$R_n = 1 + \sum_i b_{ni} r_i.$$

  Update $\{A_n\}$ to satisfy

$$\beta_n R_n = 1 \qquad \forall n.$$

until the Euler conditions converge

return $\{A_n, b_{ni}, k_i, r_i\}$.

G.2 Treatment of the Rest of the World

The set of destinations observed in the bilateral portfolio data is larger than the set of countries explicitly included in the general-equilibrium model. Dropping these destinations would mechanically reallocate their portfolio shares toward countries in the model sample. We therefore aggregate all destinations outside the model sample into a single outside option.

Let $\mathcal{N}$ denote the set of model countries and $\mathcal{R}$ the set of observed destinations outside $\mathcal{N}$. For

investor n, we construct the inclusive value

$$IV_{nR} = \log \sum_{j \in \mathcal{R}} \exp\left(-\phi_{nj} + v_j\right), \tag{G.6}$$

where v_j is the estimated destination attractiveness component. The portfolio problem then treats IV_{nR} as an additional investment alternative:

$$b_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in \mathcal{N}} \exp(V_{nj}) + \exp(IV_{nR})}, \qquad V_{ni} = -\phi_{ni} + \frac{\omega_i + r_i}{\sigma_N}. \tag{G.7}$$

Assets allocated to the outside option do not enter the capital stock of countries in $\mathcal{N}$. Hence, capital-market clearing within the model is

$$k_i = \sum_{n \in \mathcal{N}} b_{ni} A_n, \qquad i \in \mathcal{N}. \tag{G.8}$$

Because the model does not solve for an endogenous return outside the model sample, portfolio returns entering the Euler equation are computed using returns on model destinations only.

H Additional Quantitative Results

H.1 Model Fit

Figure H.1 compares the model-implied interest rates with their empirical counterparts. The model reproduces the overall cross-country variation in interest rates.

H.2 Spatial Decomposition of Destination Attractiveness

Figure H.2 presents the estimated persistent components of destination attractiveness. Panel (a) shows the spatial component, $\hat{s}_i$, while Panel (b) shows the non-spatial country-specific component, $\hat{a}_i$. Both components are recovered using the full-sample BLUP estimates described in Section 6.2.

H.3 Counterfactual Reallocation with Fixed Asset Stocks

Figure H.3 shows how capital is reallocated across destinations when each geographic channel is removed while country-level asset stocks are held fixed at their baseline values. Conditional on these

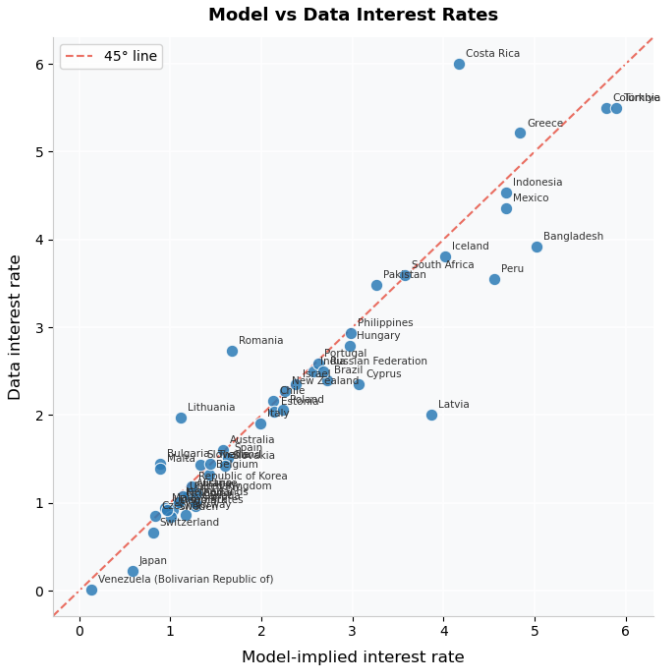


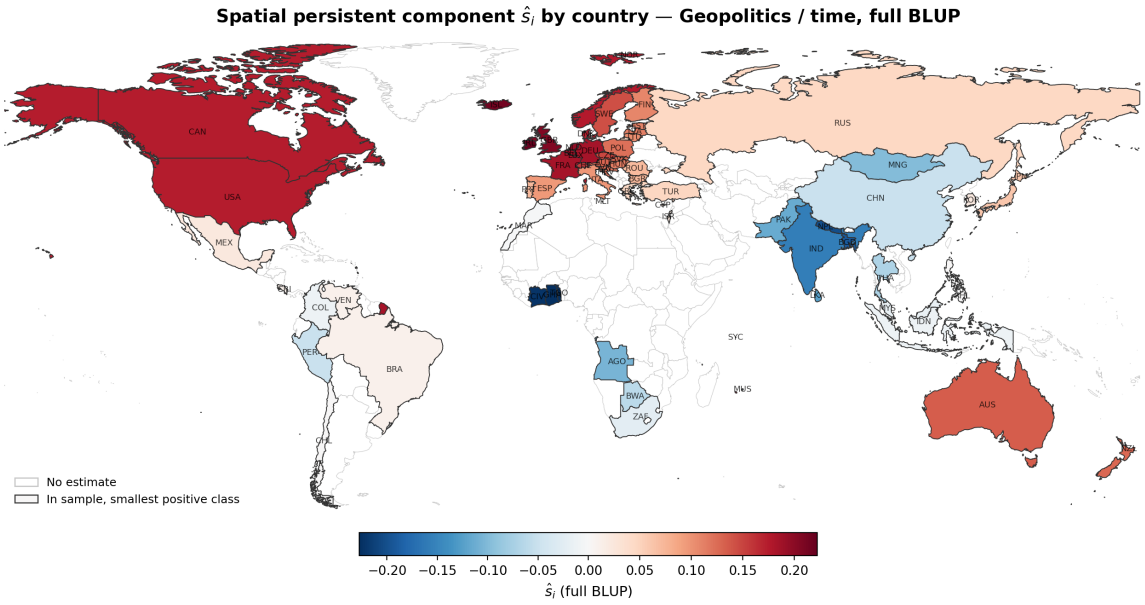
Figure H.1: Model-Implied and Observed Interest Rates

asset stocks, portfolio shares, destination capital stocks, and equilibrium interest rates are jointly re-solved.

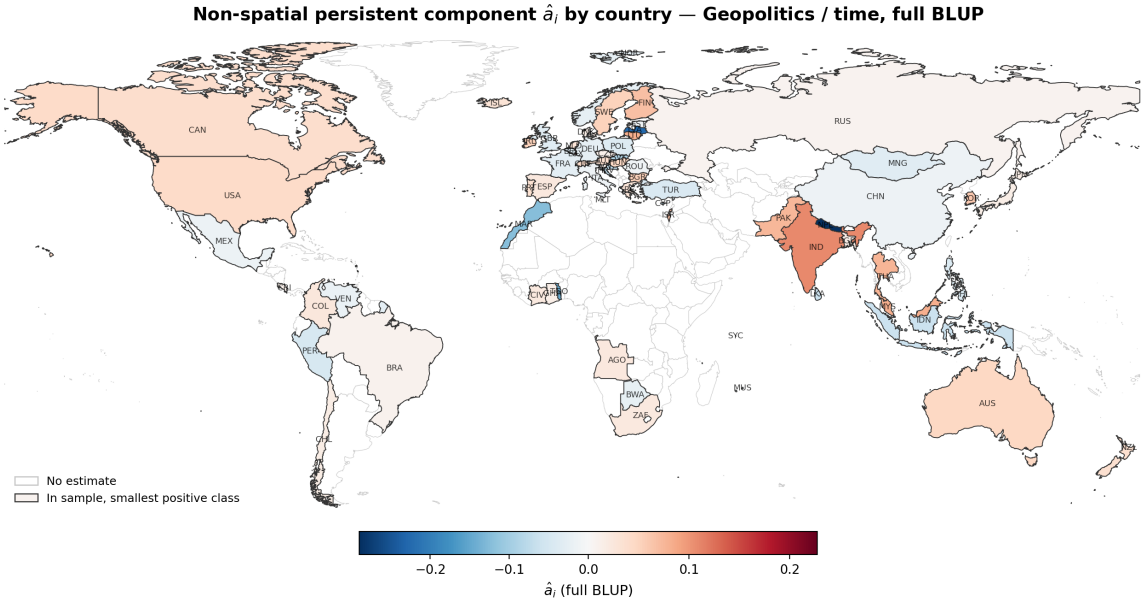
Figure H.4 reports the corresponding changes in interest rates. Each point compares the baseline interest rate with the counterfactual interest rate for the same country. Points above the 45-degree line indicate an increase in the equilibrium interest rate relative to the baseline, while points below the line indicate a decrease.

H.4 Geographic Transmission of Attractiveness Shocks

Figure H.5 illustrates how interest-rate responses vary with geographic distance between the shock origin and recipient countries.



(a) Spatial persistent component, $\hat{s}_i$



(b) Non-spatial persistent component, $\hat{a}_i$

Figure H.2: Decomposition of Destination Attractiveness

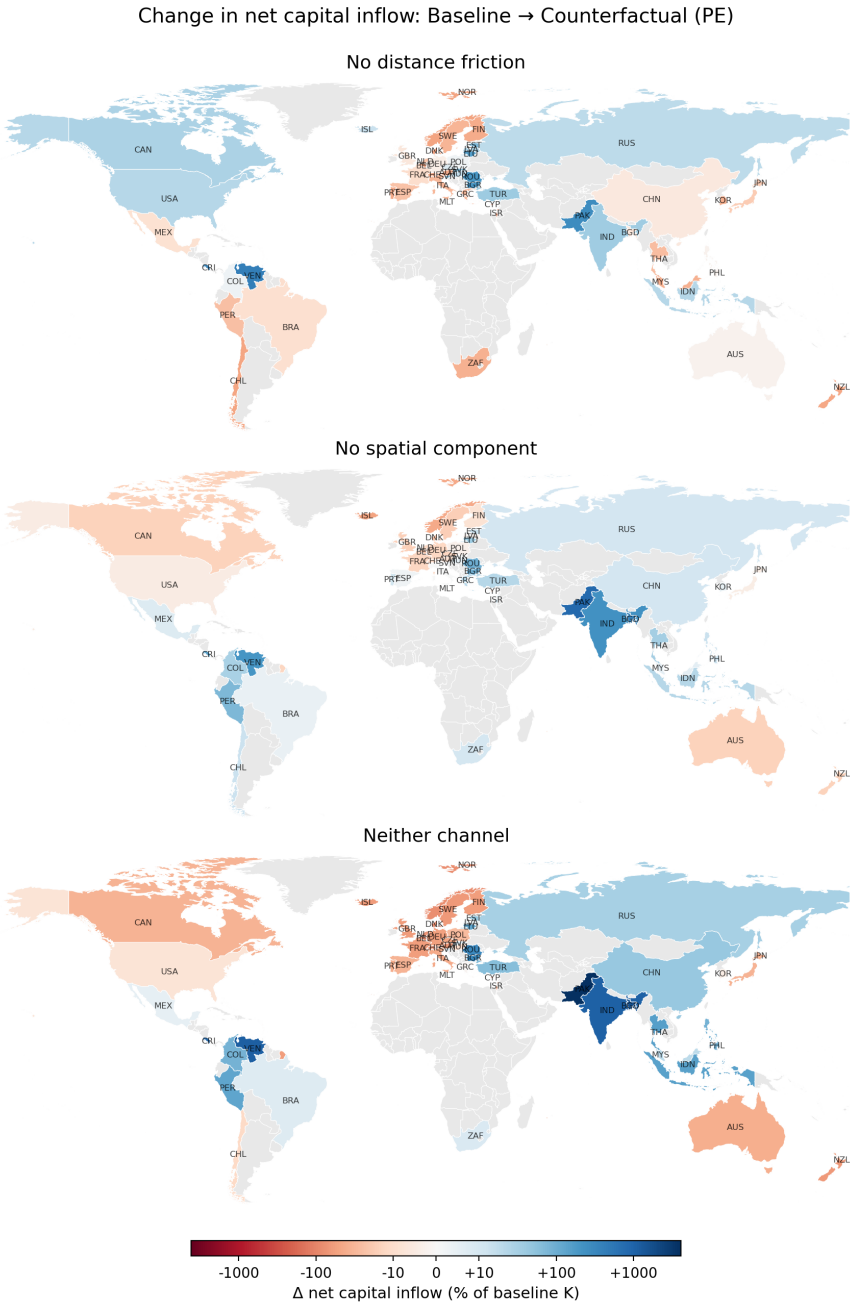


Figure H.3: Capital Reallocation with Fixed Asset Stocks

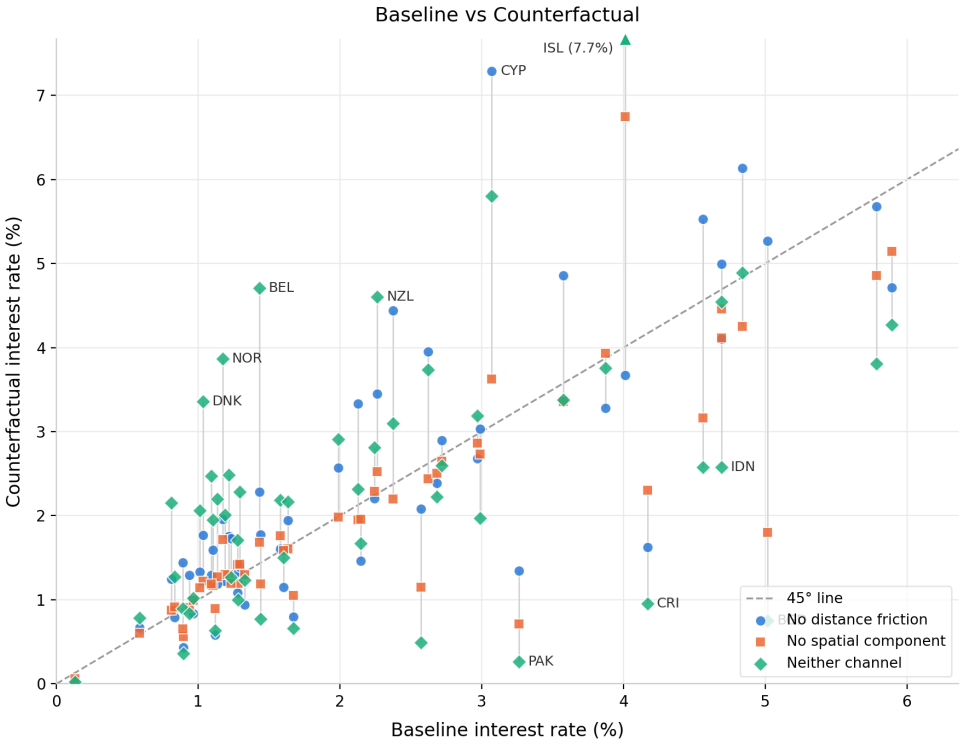


Figure H.4: Interest Rate Responses with Fixed Asset Stocks

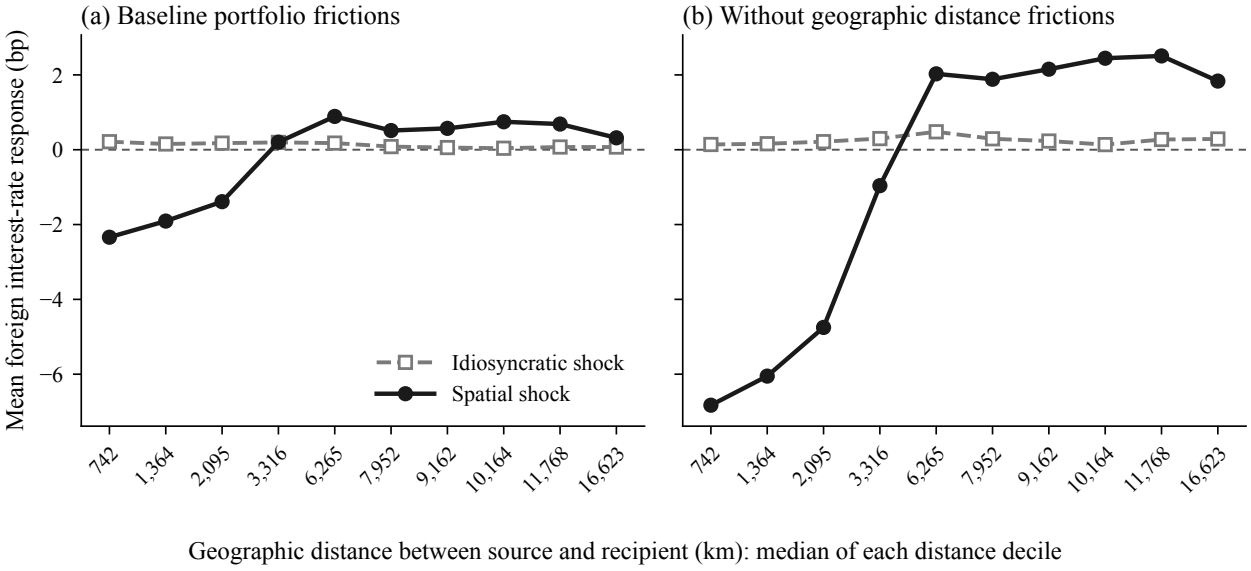


Figure H.5: Interest-Rate Responses by Geographic Distance

Notes: The figure plots mean foreign interest-rate responses across deciles of bilateral geographic distance. We consider each of the 53 countries as a shock origin and compute responses for all 2,756 directed source-recipient country pairs. Shocks are normalized to a one-percentage-point increase in return-equivalent attractiveness in the origin country. Responses are based on local general-equilibrium comparative statics and expressed in basis points. Panel (a) reports results under baseline portfolio frictions, while Panel (b) removes the geographic distance component of bilateral portfolio frictions.

I Proofs

I.1 Proof of Proposition 4.5

Proof. Consider an interior equilibrium with fixed investor asset stocks $\{A_n\}$, strictly positive portfolio shares, and decreasing marginal products of capital.

Equilibrium Interest-Rate Transmission. Let $\boldsymbol{k}$ denote the vector of destination capital stocks. The equilibrium capital-market clearing condition is

$$\boldsymbol{k} = \boldsymbol{S}(\boldsymbol{r} + \boldsymbol{\omega}), \tag{I.1}$$

where $\boldsymbol{S}(\cdot)$ denotes aggregate capital demand across investment destinations implied by portfolio choices.

Totally differentiating the market-clearing condition gives

$$d\boldsymbol{k} = J(d\boldsymbol{r} + d\boldsymbol{\omega}), \tag{I.2}$$

where

$$J = \frac{\partial \boldsymbol{S}}{\partial(\boldsymbol{r} + \boldsymbol{\omega})} \tag{I.3}$$

is the Jacobian of aggregate portfolio demand.

From the firms' optimality conditions, equilibrium interest rates are decreasing in destination capital stocks:

$$d\boldsymbol{r} = -H\, d\boldsymbol{k}, \tag{I.4}$$

where

$$H = \operatorname{diag}\left(-\frac{\partial r_i}{\partial k_i}\right)_{i=1}^{N} \tag{I.5}$$

is a positive diagonal matrix.

Substituting the capital-market clearing condition yields

$$d\boldsymbol{r} = -HJ(d\boldsymbol{r} + d\boldsymbol{\omega}). \tag{I.6}$$

Rearranging,

$$(H^{-1} + J)d\boldsymbol{r} = -J\, d\boldsymbol{\omega}. \tag{I.7}$$

Therefore,

$$d\boldsymbol{r} = -P\, d\boldsymbol{\omega}, \qquad P = (H^{-1} + J)^{-1} J. \tag{I.8}$$

This establishes the general equilibrium interest-rate transmission relationship.

Properties of the Transmission Matrix. Under the logit portfolio allocation rule, aggregate capital demand for destination i is

$$S_i = \sum_n A_n b_{ni}. \tag{I.9}$$

The elements of the Jacobian J are

$$J_{ij} = \frac{1}{\sigma_N} \sum_n A_n b_{ni} \left(\mathbb{1}[i = j] - b_{nj} \right). \tag{I.10}$$

Since portfolio shares are strictly positive,

$$J_{ii} > 0, \qquad J_{ij} < 0 \quad (i \neq j). \tag{I.11}$$

Moreover, because portfolio shares sum to one,

$$J\mathbf{1} = 0. \tag{I.12}$$

Define

$$M = H^{-1} + J. \tag{I.13}$$

Since H^{-1} is positive diagonal and J is positive semidefinite with nonpositive off-diagonal elements, M is a symmetric positive definite Stieltjes matrix.

Consequently, M^{-1} is elementwise nonnegative and, under strictly positive portfolio shares, strictly positive.

Using

$$P = (H^{-1} + J)^{-1} J = I - M^{-1} H^{-1}, \tag{I.14}$$

we obtain

$$P_{ij} = -(M^{-1})_{ij}(H^{-1})_{jj} < 0, \qquad i \neq j. \tag{I.15}$$

Furthermore, since $J\mathbf{1} = 0$,

$$P\mathbf{1} = 0. \tag{I.16}$$

It follows that

$$P_{ii} = -\sum_{j \neq i} P_{ij} > 0. \tag{I.17}$$

Since

$$P_{ii} = 1 - (M^{-1})_{ii}(H^{-1})_{ii}, \tag{I.18}$$

we also have $P_{ii} < 1$.

Therefore,

$$0 < P_{ii} < 1, \qquad P_{ij} < 0 \quad (i \neq j). \tag{I.19}$$

This completes the proof.

□